

Tropical Meteorology



Chapter 2: Tropical Climatology



Objective

- *To Define Climatology*
- *To Know The Tropical Climatology*
 - ✓ Solar Radiation
 - ✓ Temperature
 - ✓ Precipitation

Climatology

- ❑ Climatology is the study of the long-term state of the atmosphere.
- ❑ It fundamentally concerned with the weather and climate of a given area.
- ❑ It examines both the nature of micro (local) and macro (global) climates and the natural and anthropogenic influences on them.

Cont.....

- ❑ The term climate implies an average or long term record of weather conditions at a certain region for at least 30 years.
- ❑ Tropical Climatology aims to provide a geographical view point on the physical processes in the tropical atmosphere.

Cont.....

- Climate is the expected mean and variability of the weather conditions for a particular location, season, and time of day.
- *The climate* is often described in terms of the mean values of meteorological variables.
- Climatology is also concerned with *seasonal to inter-annual climate variability*, *decadal to millennial climate fluctuations*, *long-term changes in mean and variability characteristics*, climate extremes and seasonality.

Cont.....

- Tropical climatology encompasses the study of weather patterns, climate phenomena, and atmospheric dynamics in tropical regions, typically within the latitudinal band between the Tropic of Cancer and the Tropic of Capricorn.

Radiation and Energy Balance in the Tropics

Solar Radiation

The tropics receive:

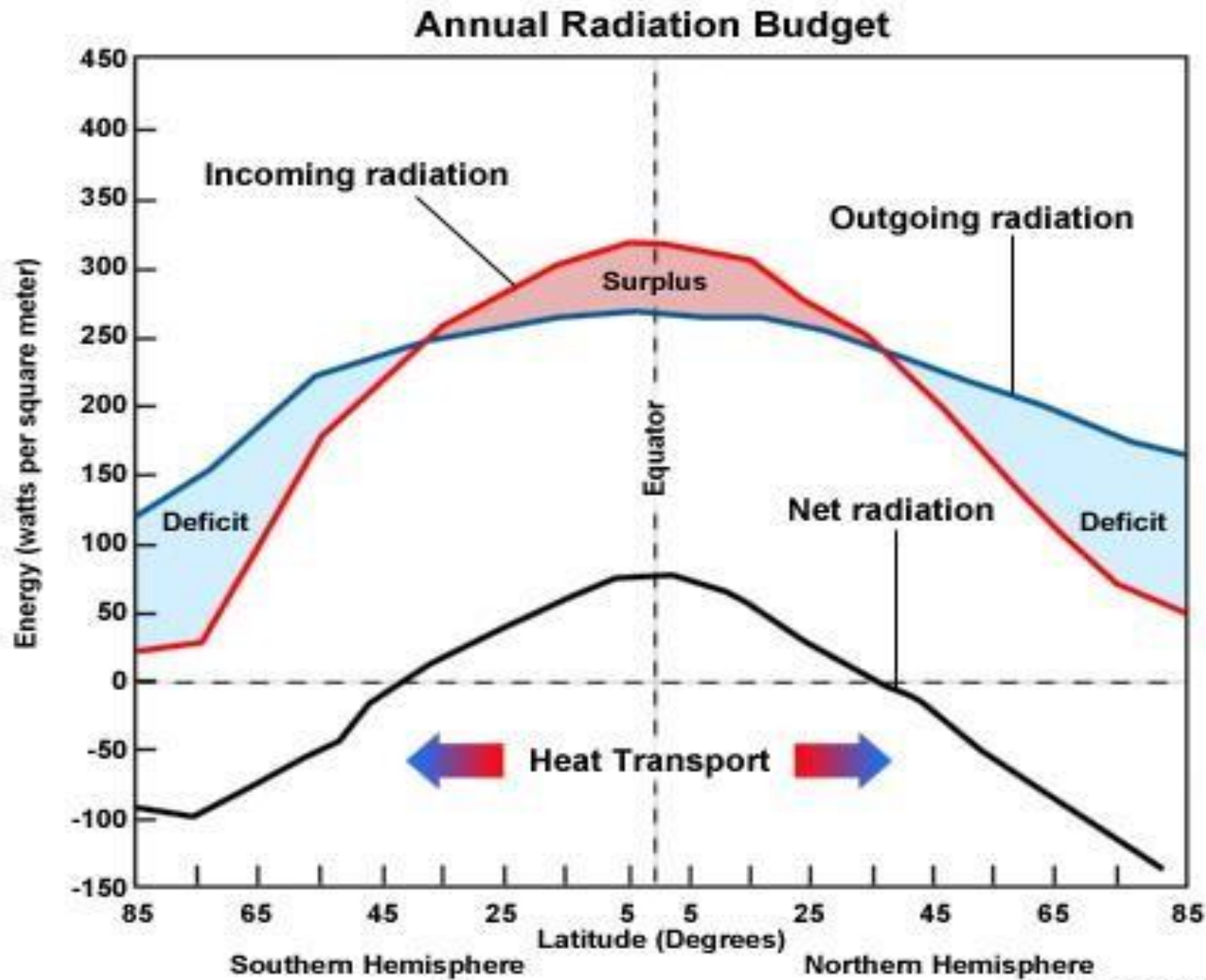
- Highest annual mean insolation
- Small seasonal variation (compared to mid-latitudes)

At the Equator:

- Day length $\approx$ 12 hours year-round
- High net radiation.

This produces a net energy surplus

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Cont.....

Tropical Energy Budget

In the global energy balance:

- Tropics: net radiation surplus
- Polar regions: net radiation deficit

The atmosphere and ocean transport energy poleward to compensate.

This is achieved mainly by:

- Hadley circulation
- Ocean currents
- Latent heat transport

Temperature

Tropical Climatology

- In general, the mean temperature declines from equator to poles and the range increases.
- Longitudinal temperature contrasts are relatively small ($< 6^{\circ}\text{C}$), and are correlated with *continents*, which are colder than oceans in winter and warmer during the summer.
- The tropics have the lowest annual temperature range for the globe, less than 1.5°C in the near equatorial regions.
- The maximum annual temperature range in Africa occurs in northwestern Africa, south of the Atlas Mountains.

*Major influences on **Annual Surface** Temperature Distribution*

Diurnal Temperature Variability in the Tropics

-  The diurnal variation in temperature has its maximum at the surface, following the daily cycle of surplus heating.
-  *The diurnal cycle of surface temperature hits its minimum temperature at sunrise and maximum in the afternoon.*
-  The daily maximum generally lags the solar maximum as the heated surface is warming the surrounding air.

Horizontal and vertical distribution of temperature

- The average surface temperature of the Earth is about 288K or 15°C.
- The global average precipitation is about 1 meter per year.
- The climate of a given location varies with

Latitude

Altitude and

geographical conditions

- Seasonal variations are greater in higher latitudes and in continental rather than maritime areas.

Cont.....

- Temperature and rainfall vary widely by locality. Near the equator, the range of temperatures is small and rainfall abundant. Away from the equator, temperatures and rainfall vary considerably.

Temperature Distribution In The Tropics

- A special feature of the latitudinal receipt of radiation is that the maximum temperatures experienced at the earth's *surface do not occur at the equator*, as one might expect, but at the tropics.
- The apparent migration of the vertical sun is relatively rapid during its passage over the equator, but its rate slows down as it reaches the tropics.
- Between 6°N and 6°S the sun's rays remain almost vertically overhead for only 30 days during each of the spring and autumn equinoxes, allowing little time for any large buildup of surface heat and high temperatures.

Cont.....

- On the other hand, between 17.5°N and 23.5°N latitude the sun's rays shine down almost vertically for 86 consecutive days during the period of the solstice.
- This longer interval, combined with the fact that the tropics experience longer days than at the equator, makes the maximum zones of heating occur nearer the tropics than the equator.

Cont.....

- In the Northern Hemisphere, this poleward displacement of the zone of maximum heating is enhanced by the effect of continentality, while low cloudiness associated with the subtropical high pressure belts is an additional factor.
- The clear skies allow large annual receipts of solar radiation in these areas.
- Over land, the highest values (38–40°C) occur at about 23°N and 10–15°S.

Highest Temperatures in the Tropics

The hottest tropical regions are usually

:

- ✓ Subtropical deserts near 20–30° latitude
- ✓ Dry continental interiors
- ✓ Low humidity → strong sensible heating

Sahara Desert (Northern Tropics Edge)

- ❑ The Sahara Desert is the world's largest hot desert, spanning much of North Africa.
- ❑ Covering about 9 million square kilometers (3.5 million square miles), it stretches from the Atlantic Ocean in the west to the Red Sea in the east, forming a vast arid expanse that divides the African continent into northern and sub-Saharan regions.
- ❑ Southern Sahara lies within the tropics (around 15–23°N).



Why Sahara Desert so hot?

- ❖ Strong solar radiation
- ❖ Clear skies
- ❖ Very low soil moisture
- ❖ Low latent heat flux (LE small)
- ❖ Large sensible heat flux (H large)
- ❖ So surface temperature rises dramatically

Lowest Temperatures in the Tropics

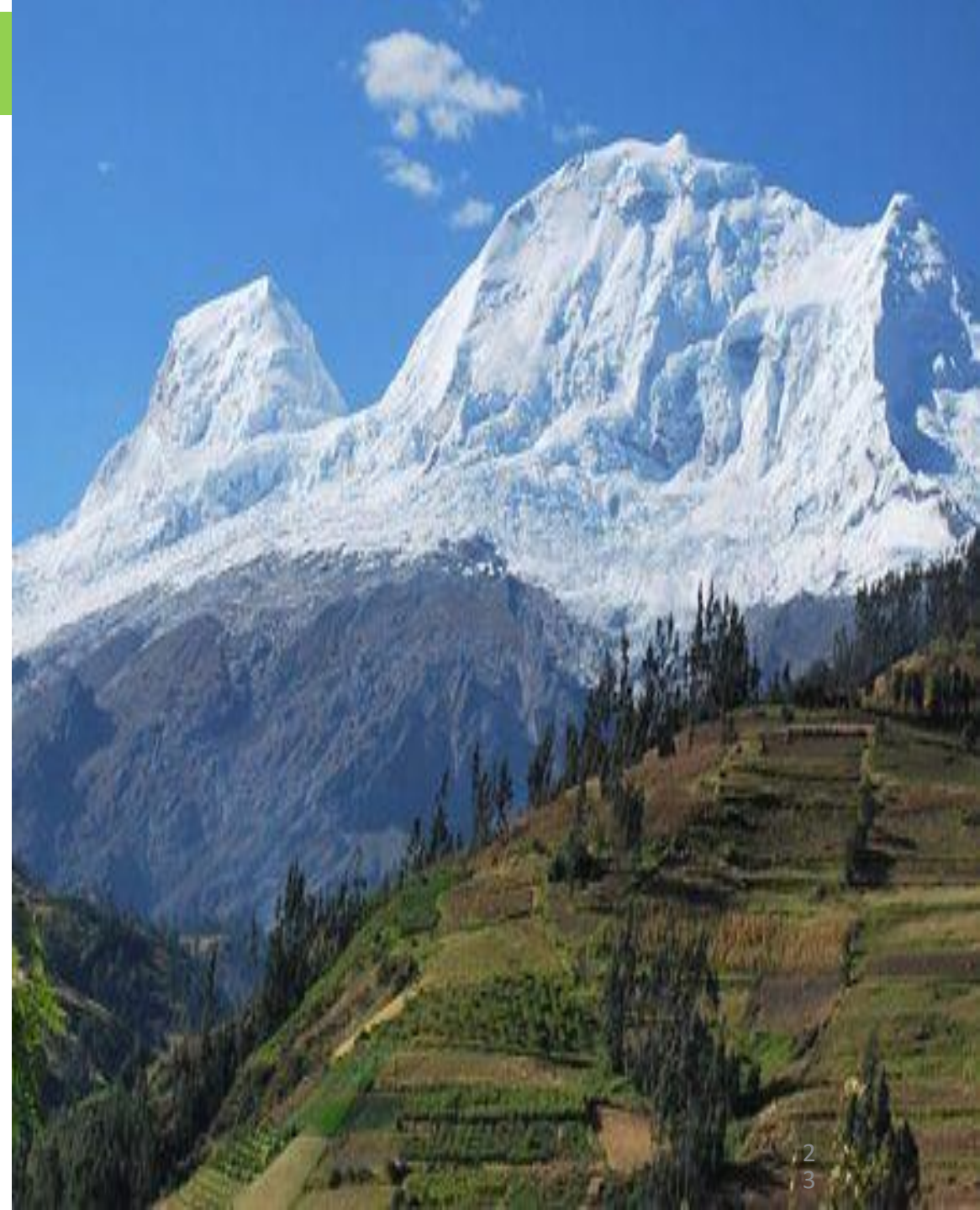
Chimborazo (Ecuador)

- Height: 6,263 m
- Latitude: $\sim 1^{\circ}\text{S}$
- Why very cold?
 - Highest mountain in the tropics
 - Permanent glaciers (though retreating)
 - Temperatures often below -10°C at the



Huascarán (Peru)

- Height: 6,768 m
- Latitude: $\sim 9^{\circ}\text{S}$
- Why very cold?
 - Highest mountain in tropical Andes
 - Heavy glaciation
 - Summit temperatures frequently below -15°C



Nevado Sajama (Bolivia)

- Height: 6,542 m
- Latitude: $\sim 18^{\circ}\text{S}$
- Why very cold?
 - High elevation + dry air of Altiplano
 - Night temperatures extremely low



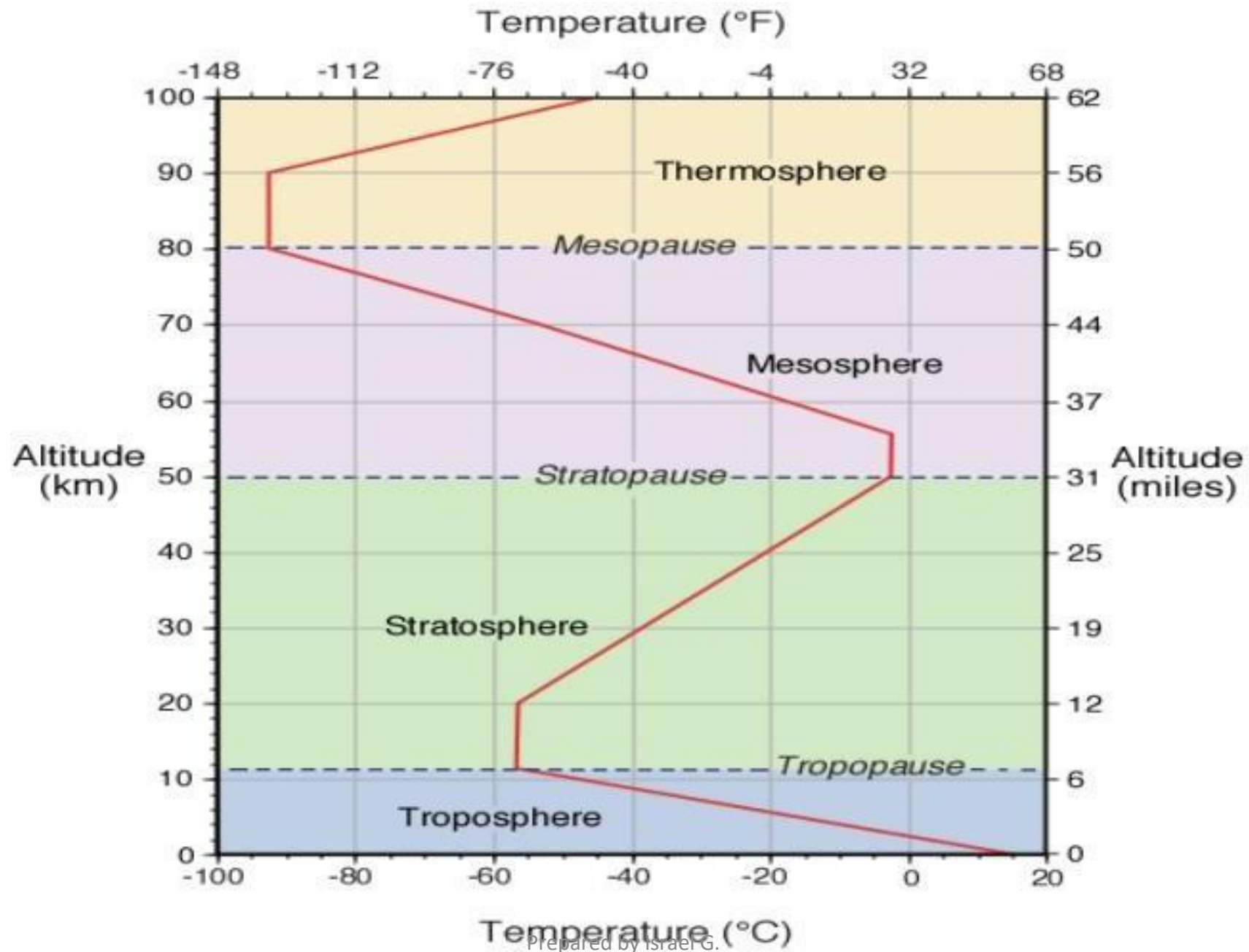
Mount Kilimanjaro (Tanzania)

- Height: 5,895 m
- Latitude: $\sim 3^{\circ}\text{S}$
- Why cold?
 - High elevation near equator
 - Glacier remnants at summit
 - Night temperatures below -20°C



Vertical Distribution Of Temperature

- In the standard atmosphere the temperature in the troposphere decreases at a constant rate of $6^{\circ}\text{C}/\text{km}$ up to 11 km and then temperature remains constant up to 11 to 20 km called tropopause.
- The stratosphere consists of two sub layers. The increase of temperature from 20 to 47 km and then remains constant from 47-52 km called stratopause.
- Later the decrease of temperature from 52-79 km is called mesosphere and the mesopause is between 80-90km.
- The thermosphere begins at 90 km and then temperature increases again continuously.



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- The troposphere varies between equator and the poles.
- At the equator it is at higher elevation (about 16 km) and at poles it is at lower elevation (8 km).
- This is because as the lithosphere (solid part of the earth) bulges at the equatorial latitudes and shrinks at the poles, the atmosphere also takes the same shape due to rotational inertia of the earth.
- The minimum temperatures at the tropopause at the equator and poles respectively are -62°C and -45°C .

Temperature Controls Atmospheric Thickness

$$Z_2 - Z_1 = \frac{RT_v}{g} \ln \left(\frac{P_1}{P_2} \right)$$

where:

- $Z_2 - Z_1$ = thickness of the atmospheric layer
- R = specific gas constant for dry air
- T_v = virtual temperature (accounts for water vapor)
- g = gravitational acceleration
- P_1, P_2 = pressure at different heights

From the hypsometric equation:

❑ The thickness of an atmospheric layer is proportional to its mean temperature.

- Warm air → expands → thicker troposphere
- Cold air → contracts → thinner troposphere

❑ At the equator:

- Strong solar radiation
- High surface temperatures
- Deep convective heating

Result → Expanded atmospheric column → Tropopause ~16–17 km

Cont.....

❑ At the poles:

- Weak solar radiation
- Very cold air masses

Result → Compressed atmospheric column → Tropopause ~8km

Cont.....

- One point may be noted here while temperatures decrease with height in the troposphere, temperatures increase with height in the stratosphere.
- The reasons for this are that while the troposphere is influenced by the earth's surface from below, the stratosphere is influenced by sun's radiation from above.
- The troposphere is largely heated through the long wave radiation from the earth's surface through the absorption of CO₂ and water vapor.
- Where as the stratosphere is largely heated through the absorption of UV radiation by ozone from the sun which means it is heated from above

Long-term temperature trends and variability in the Tropics

- Like the rest of the world, the Tropics have warmed over the last century, with steady warming from the early 20th century to about 1940, then a levelling off until the mid-1970s, then a rapid rise to the end of the century.
- Depending on the data set used, the total warming over the 1910-2012 period in the Tropics and near-Tropics has been 0.7-0.8°C, about 0.1°C less than that over the globe as a whole.

Cont.....

- Warming over the Tropics has been slower than over higher latitudes of the Northern Hemisphere, but faster than over mid- to high latitudes of the Southern Hemisphere.
- In the last 30 years, the Sahara, Sahel and Arabian Peninsula have been amongst the most rapidly warming parts of the world, whilst large parts of the eastern and central tropical Pacific have seen weak cooling, associated with decadal variability in the El Niño-Southern Oscillation (ENSO).

Cont.....

- Temperatures averaged over the Tropics show more inter annual variability than those averaged over the globe as a whole.
- A strong El Niño year will typically be about 0.2°C warmer in the Tropics than the years around it, and a strong La Niña year about 0.2°C cooler.

Precipitation

Moisture and Precipitation

-  Atmospheric precipitation in tropics consists of almost entirely rainfall.
-  It is the most variable element of the tropics. It differs widely from year to year and in addition it varies strongly with place as well.
-  Hence rainfall is the main factor used in the determination and subdivision of tropical climates.

Cont.....

- ❖ Ultimately, all atmospheric precipitation is the result of upward movements of air, which cause cooling by expansion by expansion beyond the level of condensation.
- ❖ In the tropic, the main cause of rising air is ***Convection*** which is responsible, alone or in combination with other factors, for most tropical rainfall.

Physical Controls of Tropical Rainfall

Solar Radiation and Convection

- Tropics receive high solar radiation year-round → strong surface heating
- Heating + abundant moisture → deep convection → heavy rainfall
- Main mechanism: convective rainfall (cumulonimbus)

Average tropical rainfall pattern (approximate)

Latitude	Typical Rainfall	Cause
0° (Equator)	Very high (~2,000–3,000 mm/year)	ITCZ, deep convection, convergence of trade winds
5–15°N & S	Moderate to high (1,500–2,000 mm/year)	Monsoon influence, secondary convection
15–30°N & S	Low (100–500 mm/year)	Subtropical highs, descending air, deserts

Wettest Tropical Regions

- ❑ Amazon Basin (South America) → ~2,000–3,000 mm/year
- ❑ Congo Basin (Central Africa) → 1,500–2,500 mm/year
- ❑ Maritime Southeast Asia (Indonesia, Malaysia) → 2,000–4,000 mm/year
- ❑ Western Pacific islands (Philippines, Papua New Guinea) → 2,000–5,000 mm/year

Moderate Rainfall Regions

❑ Tropical savannas, seasonal monsoon zones

❑ Example:

- Northern India → 800–1,200mm/year (summer monsoon)
- West Africa (Sahel) → 500–1,000 mm/year
- Northern Australia (wet season) → 1,000–1,500 mm/year

Lowest Rainfall Regions (Tropical Deserts)

❖ Annual rainfall < 250 mm/year

❖ Example:

- Sahara Desert → 50–200 mm/year
- Arabian Desert → 50–100 mm/year
- Namib Desert (Southern Africa) → <100 mm/year

Seasonal Variability

- Equatorial regions ($0-5^{\circ}$) → Rainfall almost year-round (no strong dry season)
- Monsoon regions ($5-15^{\circ}$) → Strong wet/dry season
 - Wet season aligns with solar maximum over continent
 - Dry season occurs when ITCZ moves away
- Subtropics ($15-30^{\circ}$) → Mostly dry all year, occasional summer convection

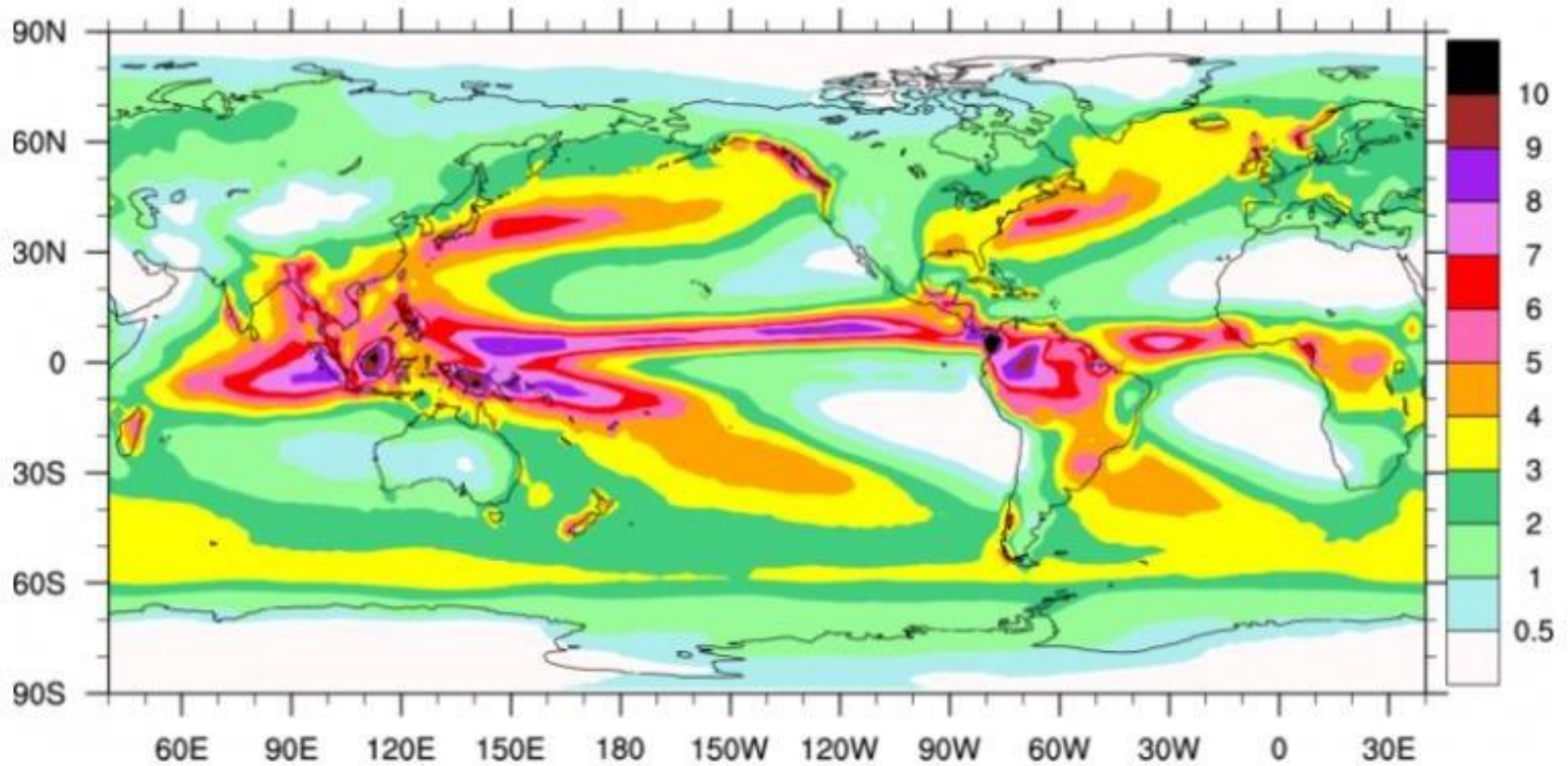


Figure 1. Annual average precipitation rate (in mm per day). This estimate is based on information from rain gauges over land and satellite measurements over the oceans.© 2010 UCAR

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Thanks !

Tropical Meteorology



Chapter 2: Tropical Climatology



Objective

- *To Know The Tropical Climatology*
 - ✓ *Air Pressure*
 - ✓ *Wind*
- *To Explain the Climate Of Tropical Africa With Particular Reference To East Africa*

Air Pressure

Air Pressure

- In atmospheric dynamics, pressure gradients drive wind through the Pressure Gradient Force (PGF).
- What kind of pressure dominates near the equator — high pressure or low pressure?

Distribution Of Air Pressure

- The weight of a column of air at a given place and time is called air pressure.
- Barometer is the instrument which measures air or atmospheric pressure.
- The unit of measurement of atmospheric pressure is millibar (kilopascals).

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- The mean atmospheric pressure at sea level is 1013.25 millibars.
- Distribution of atmospheric pressure on the surface of the earth is not uniform. It varies both vertically and horizontally.

Vertical Distribution Pressure

- Air is a mixture of various gases. It is highly compressible. As it compresses, its density increases. The higher the density of air, the greater is the air pressure and vice versa.
- The mass of air above in the column of air compresses the air under it hence its lower layers are more dense than the upper layers; As a result, the lower layers of the atmosphere have higher density, hence, exert more pressure.
- Conversely, the higher layers are less compressed and, hence, they have low density and low pressure.

Cont.....

- The columnar distribution of atmospheric pressure is known as *vertical distribution of pressure*.
- Air pressure decreases with increase in altitude but it does not always decrease at the same rate.
- Dense components of atmosphere are found in its lowest parts near the mean sea level.

Cont.....

- Temperature of the air, amount of water vapour present in the air and gravitational pull of the earth determine the air pressure of a given place and at a given time.
- Since these factors are variable with change in height, there is a variation in the rate of decrease in air pressure with increase in altitude.
- The normal rate of decrease in air pressure is 34 millibars per every 300 metres increase in altitude.

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- In high mountainous areas rice takes more time to cook because low pressure reduces the boiling point of water.
- Breathing problem such as faintness and nose bleedings are also faced by many trekkers from outside in such areas because of low pressure conditions in which the air is thin and it has low amount of oxygen content.

Horizontal Distribution Pressure

- The distribution of atmospheric pressure over the globe is known as ***horizontal distribution of pressure***.
- It is shown on maps with the help of isobars. An isobar is a line connecting points that have equal values of pressure.
- Isobars are analogous to the contour lines on a relief map.

Cont.....

- The spacing of isobars expresses the rate and direction of change in air pressure. This change in air pressure is referred to pressure gradient.
- Pressure gradient is the ratio between pressure difference and the actual horizontal distance between two points.
- Close spacing of isobars expresses steep pressure gradient(strong winds) while wide spacing indicates gentle pressure gradient.

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- The horizontal distribution of atmospheric pressure is not uniform in the world.
- It varies from time to time at a given place;
- It varies from place to place over short distances.

Pressure Belts

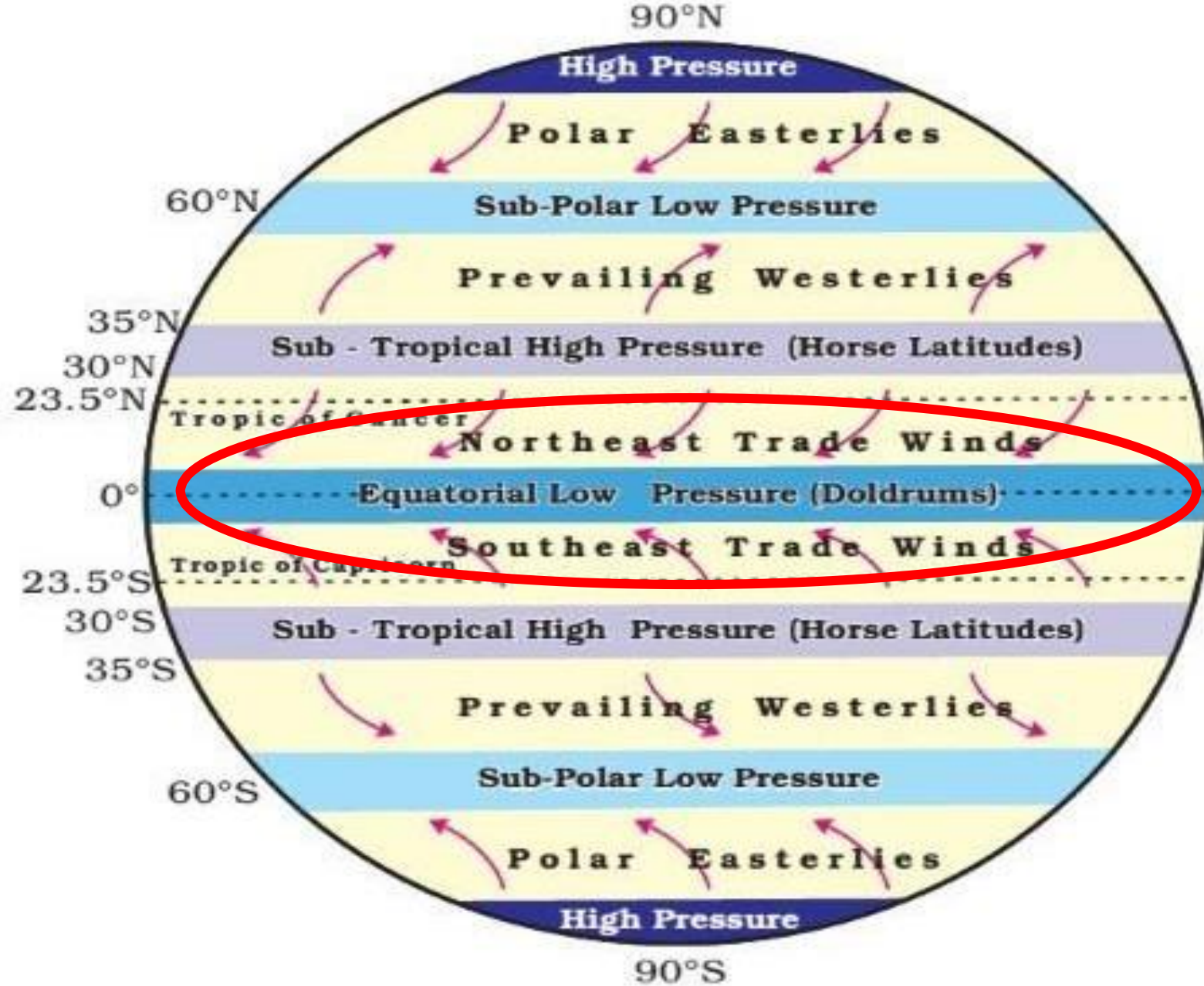
- The horizontal distribution of air pressure across the latitudes is characterized by high or low pressure belts

(i) The Equatorial Low Pressure Belt;

(ii) The Sub tropic High Pressure Belts;

(iii) The Sub-polar Low Pressure Belts;

(iv) The Polar High Pressure Belts



Major Pressure Belts and Wind System

Prepared by Israel G.

Equatorial Low Pressure Belt or 'Doldrums'

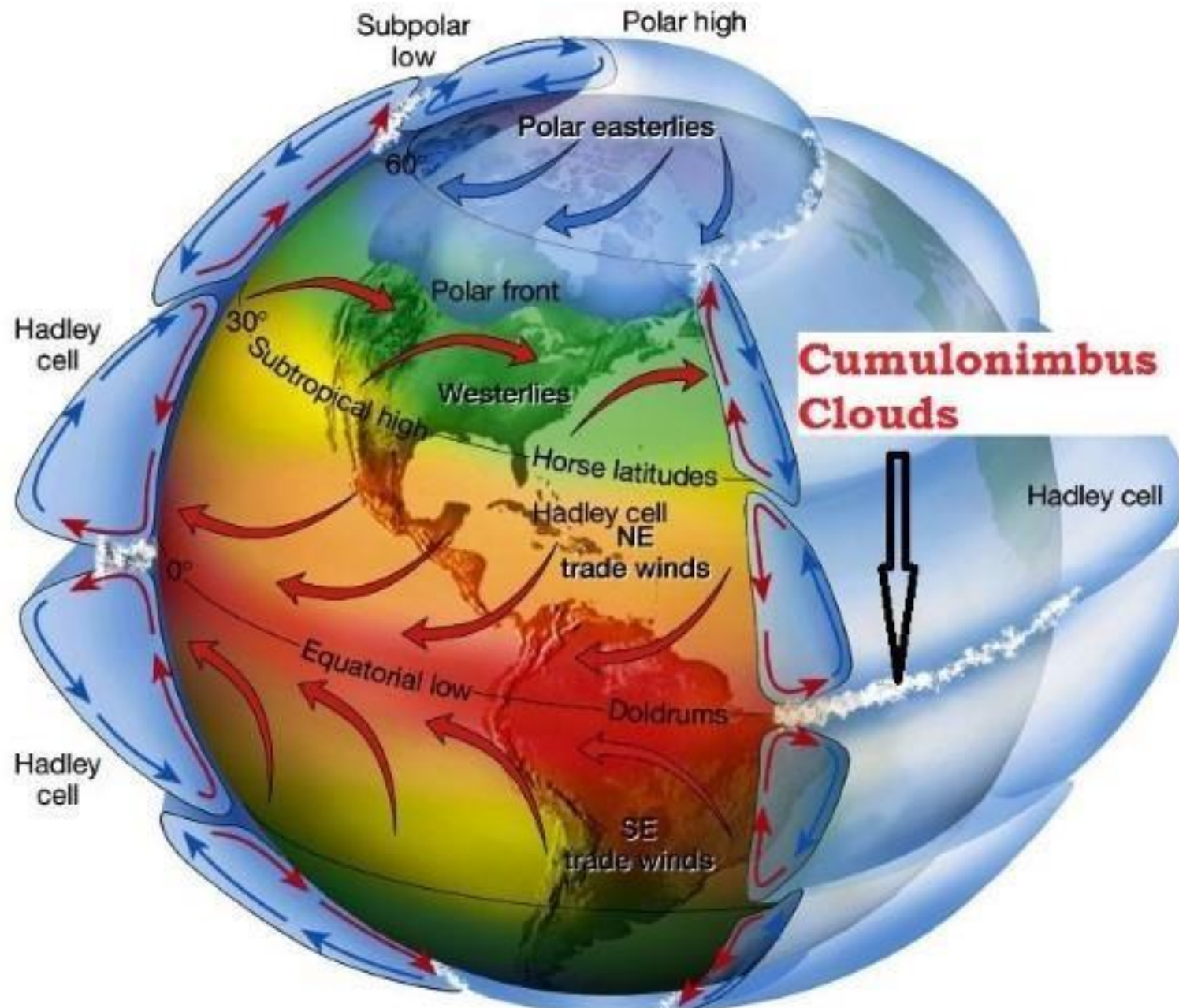
- Lies between 10°N and 10°S latitudes.
- Width may vary between 5°N and 5°S and 20°N and 20°S.
- *This belt happens to be the zone of convergence of trade winds from two hemispheres from sub-tropical high pressure belts.*
- This belt is also called the Doldrums, because of the extremely calm air movements.
- The position of the belt varies with the apparent movement of the Sun.

Formation

- As this region lies along the equator, it receives highest amount of insolation.
- Due to intense heating, air gets warmed up and rises over the equatorial region (convection).
- *Whenever there is vertically upward movement of air, the region at the surface will be at low pressure.*
- Thus the belt along the equator is called equatorial low pressure belt.

Climate

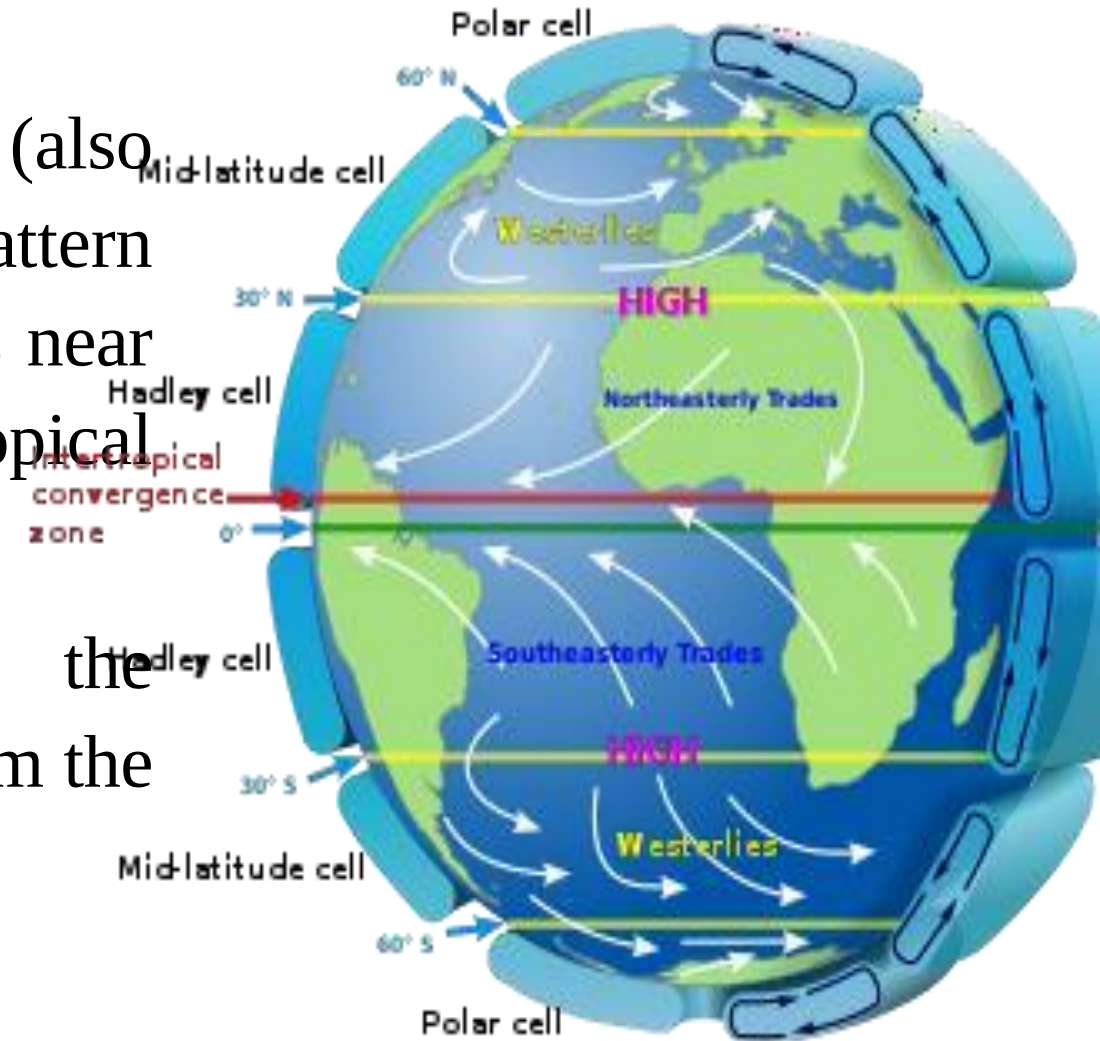
- This belt is characterized by extremely low pressure with calm conditions.
- This is because of the absence of Surface winds since winds approaching this belt begin to rise near its margin. Thus, only vertical currents are found.
- As the larger part of the low pressure belt passes along the oceans, the winds obtain huge amount of moisture.
- *Vertical winds (convection) carrying moisture form cumulonimbus clouds and lead to thunderstorms (convictional rainfall).*



Winds

Cont.....

- In general, winds are predominantly easterly at low latitudes globally.
- **Trades and their impact** : The trade winds (also called trades) are the prevailing pattern of easterly surface winds found in the tropics near the Earth's equator, equator ward of the subtropical ridge.
- These winds blow predominantly from the northeast in the Northern Hemisphere and from the southeast in the Southern Hemisphere



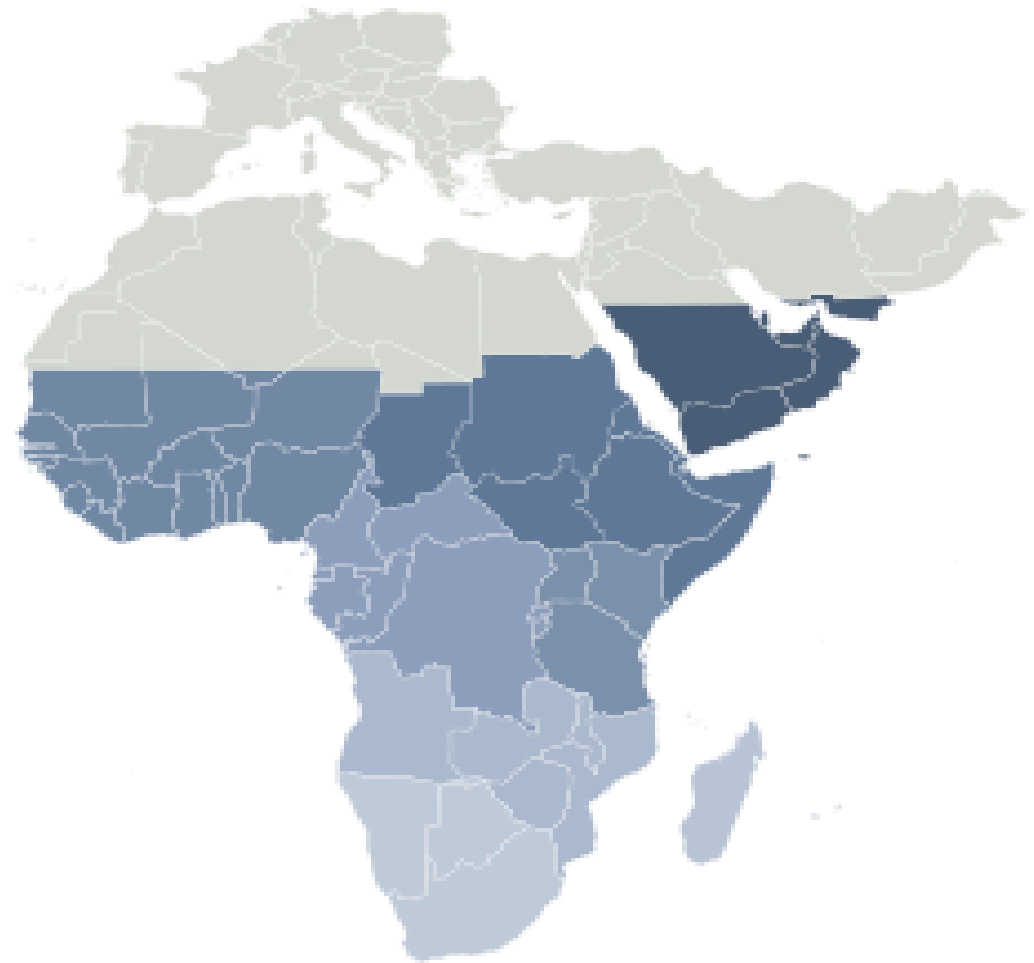
Climate Of Tropical Africa With Particular Reference To East Africa

Cont.....

- Africa, with the equator cutting across its center, is the world's most tropical continent.
- Only its northern edge and southern tip are outside the tropics. Half of Africa lies north of the equator and half to the south.
- This symmetry, or balance, produces matching belts of climate at approximately equal distances north and south of the equator.

Cont.....

- *The center of the continent has a wet tropical climate, with extremely heavy rainfall.*
- To the north and south are belts of tropical climate with a dry season. Beyond lie belts of tropical climate with longer dry periods and occasional droughts, such as those that have caused famines in the SAHEL.
- North of the Sahel lies the almost rainless SAHARA DESERT.



Cont.....

- *Equatorial Africa includes the southern coast of West Africa and the rainforests of Central Africa.*
- *This region receives heavy rain throughout the year, although there are two periods with even more rainfall.*
- These rainier seasons generally last from *September to November* and from *February to June*, although they are longer near the coast.

Cont.....

- The interior of West Africa, the Sahel, Sudan, and the Ethiopian highlands form another, drier climate region.
- The northern parts of this area receive less precipitation than the southern parts and have a single rainy season.
- In addition, rainfall in the region generally decreases from west to east. Parts of eastern Ethiopia and Somalia near the Indian Ocean have a desert climate.

Cont.....

- East Africa, consisting of Uganda, Kenya, and Tanzania, straddles the equator.
- It has a hot, humid climate along the coast and around Lake Victoria, but cooler climates in the highlands and mountains.
- The rainiest areas are the Lake Victoria basin, the mountains, and the coastal islands, which receive more than 59 inches of rain annually.
- *Generally the East African climate is characterized by high temperatures and low rainfall*

Köppen climate classification system

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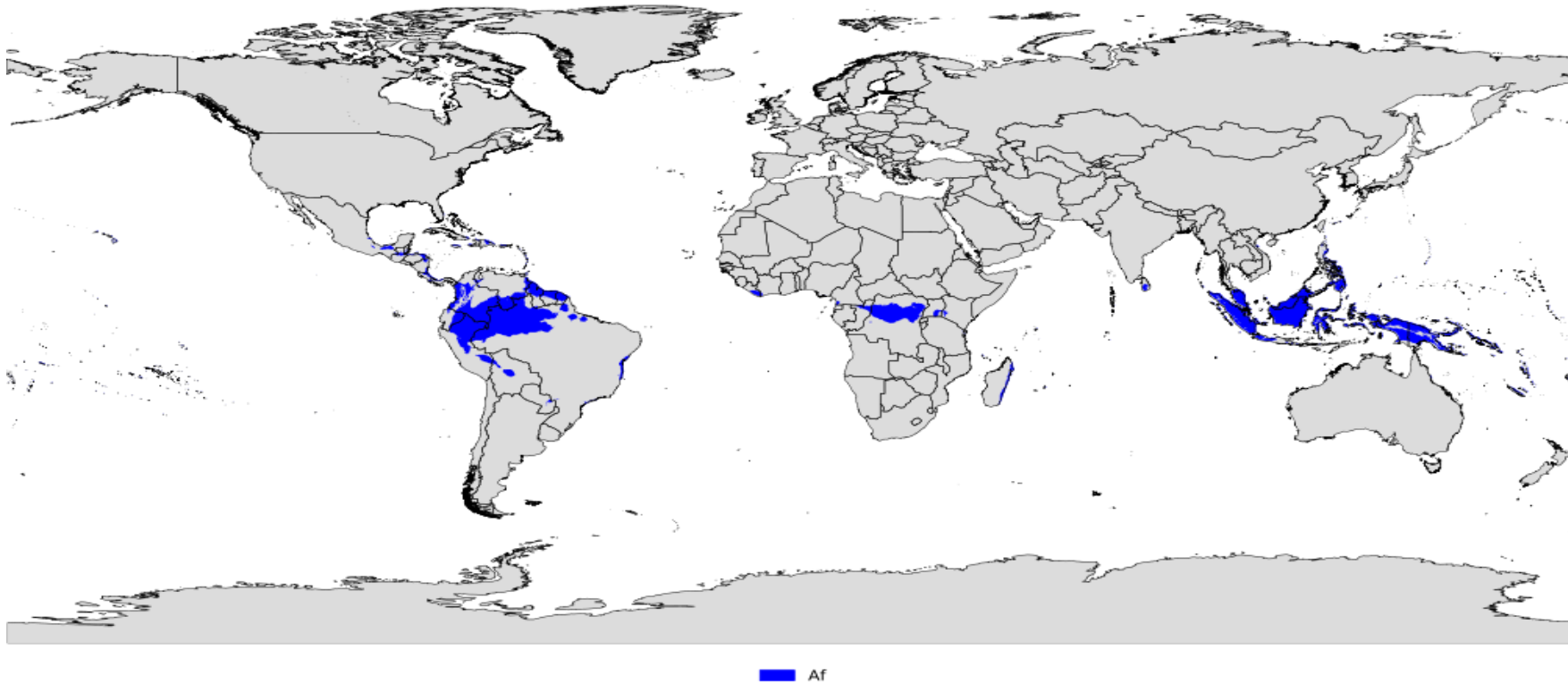
- Tropical climates are generally divided into several subcategories based on factors such as temperature, precipitation, and vegetation.
- the Köppen climate classification system, which takes into account *temperature and precipitation patterns*. Each subtype has its unique characteristics, influencing the local ecosystems and human activities in the region.

Tropical Rainforest Climate (Af) or equatorial climate

- usually found within 10 to 15 degrees latitude of the equator
- They experience high mean annual temperatures, small temperature ranges, and rain that falls throughout the year.
- Regions with this climate are typically designated Af by the Köppen climate classification.
- A tropical rainforest climate is typically hot, very humid, and wet with no dry season.
- These areas are lush with dense vegetation.

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Tropical, rainforest (Af) (1980–2016)



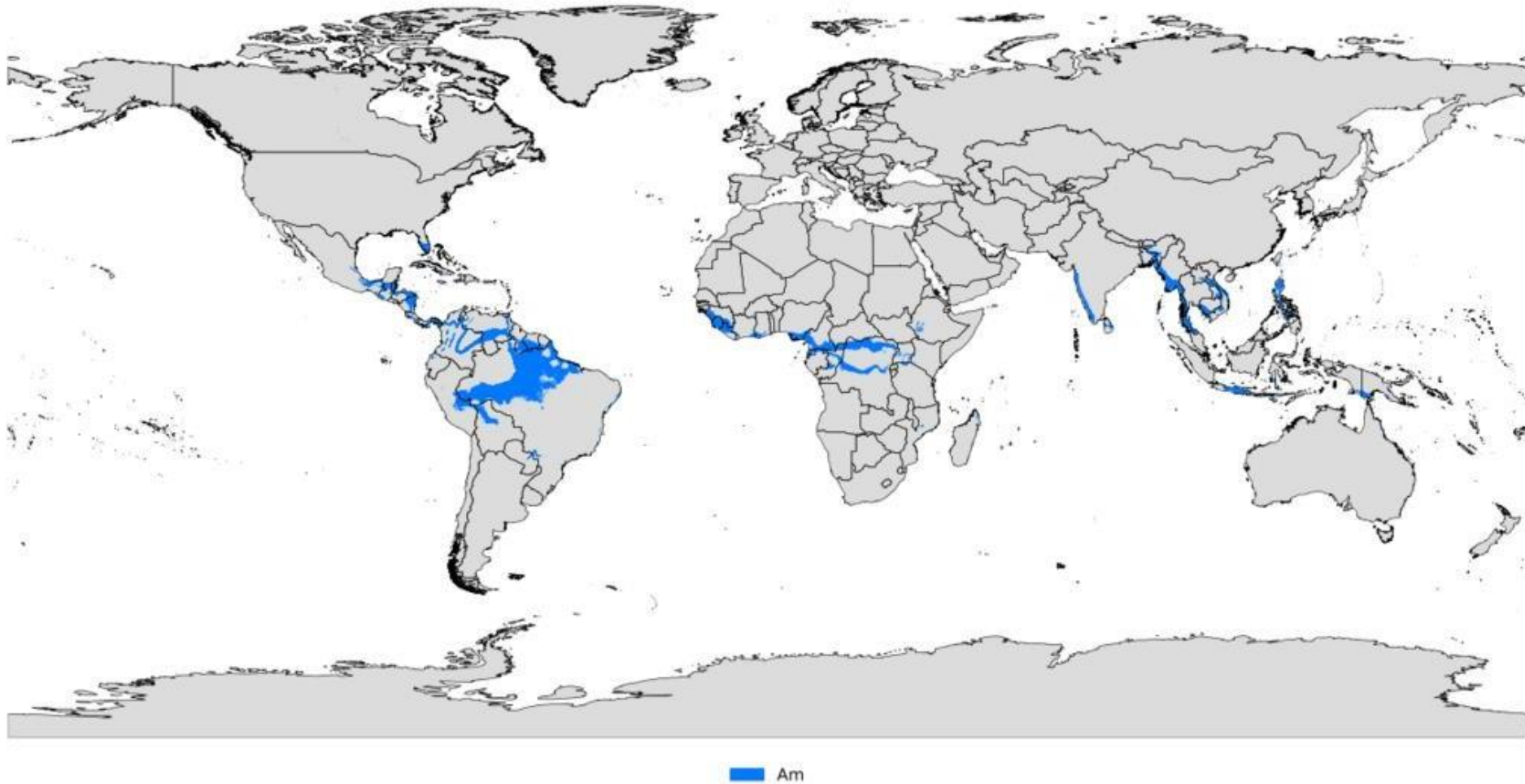
Source: Beck et al.: Present and future Köppen-Geiger climate classification maps at 1-km resolution, Scientific Data 5:180214, doi:10.1038/sdata.2018.214 (2018)

Tropical Monsoon Climate (Am)

- Similar to tropical rainforests but with distinct wet and dry seasons.
- Rainfall is heavy during the wet season, while the dry season experiences significantly less precipitation.
- The tropical monsoon climate is the intermediate climate between the wet AF (or tropical rainforest climate) and the drier Aw (or tropical savanna climate).

Cont.....

Tropical, monsoon (Am) (1980-2016)

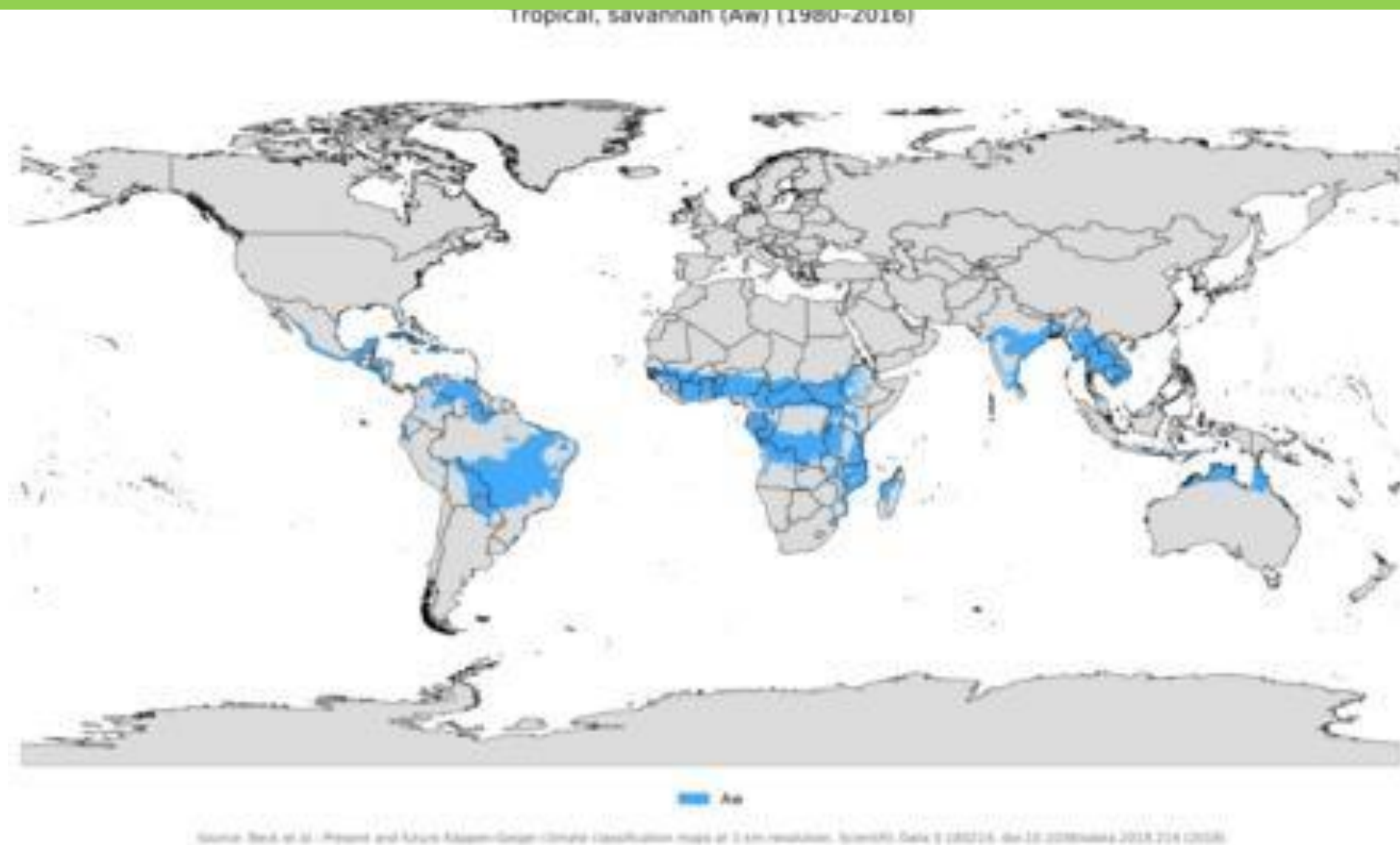


Source: Beck et al.: Present and future Köppen-Geiger climate classification maps at 1-km resolution, Scientific Data 5:180214, doi:10.1038/sdata.2018.214 (2018)

Tropical savanna climate or tropical wet and dry climate (Aw)

- *is a type of climate that corresponds to the Köppen climate classification categories Aw (for a dry winter) and As (for a dry summer, which is unusual).*
- Typically found on the outskirts of tropical rainforests, characterized by distinct wet and dry seasons. Rainfall is moderate, and there is a noticeable dry season where grasslands dominate the landscape.

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Cont.....

- Understanding tropical climatology is essential for various reasons, including agriculture, water resource management, disaster preparedness, and conservation efforts in tropical regions.
- Additionally, with climate change, studying tropical climatology becomes even more critical to understand how these regions may be affected by shifting weather patterns and rising temperatures.

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Thanks !